The realistic non-hydrostatic model nh_realistic

nh_realistic is pe_realistic with a prognostic vertical velocity and a non-hydrostatic pressure. This text describes only the differences to pe_realistic. Everything that is not mentioned here is as described in the documentation of pe_realistic and pe_simple.

The differences are the following. The vertical velocity w has its own momentum equation. A three-dimensional pressure p_{nh} makes the velocity free of divergence in every cell. The horizontal coordinates must be Cartesian.

The package contains only what is new. The grid, the masks, the tracers, the equation of state, the vertical mixing, the TKE closure, the isoneutral mixing, the surface pressure and the split-explicit external mode are imported from pe_realistic. The file `momentum.py` holds the equation for w. The file `vmix.py` holds the vertical friction of w. The files `pressure.py` and `solver.py` hold the equation for p_{nh} and its solver. The file `model.py` holds the time step.

The formulation follows pyOM2 with `enable_hydrostatic` off and the non-hydrostatic model of Oceananigans. The solver for p_{nh} follows Oceananigans. The tests and their results are listed in the section on the nhr family of `pyOM3/test/README_test_suite.md`.

Grid

The grid is that of pe_realistic with Cartesian coordinates. The widths `dxt`, `dyt` and `dzt` are in metres and may vary in all three directions. The model stops with an error if `coord_degree` is set. On the sphere the equation for w would need the metric term $-(u^2+v^2)/a$ and the Coriolis terms with $\cos\phi$. nh_realistic has neither of them.

The metric factors `cost` and `cosu` of pe_realistic are therefore 1, and `tantr` is 0. The kernels imported from pe_realistic still contain them. The new kernels of nh_realistic do not. The horizontal area of a T cell is $A^t = \Delta x^t \Delta y^t$, and its volume is $V^t = A^t \Delta z^t$.

The vertical velocity w_k sits at the top face of the T cell k. This face is the centre of the W cell k. The W cell k reaches from the centre z^t_k to the centre z^t_{k+1}. Its height is Δz^w_k, and its volume is $A^t \Delta z^w_k$.

The mask `maskW` of pe_realistic is open at the surface face, since the free surface needs a vertical velocity there. nh_realistic adds the mask `maskWc`, which is `maskW` with the surface face closed. The equation for w, its friction and the equation for p_{nh} use `maskWc`. The surface face thus belongs to the free surface and not to the equation for w.

Pressure

All pressures are divided by ρ_0, as in pe_realistic. The pressure is the sum

$$p = p_h + p_s + p_{nh} \tag{1}$$

of the hydrostatic pressure p_h, the surface pressure p_s and the non-hydrostatic pressure p_{nh}. p_h and p_s are computed as in pe_realistic.

The momentum equations are

$$\partial_t \boldsymbol{u} + \ldots = -\boldsymbol{\nabla}_h \left(p_h + p_s + p_{nh}\right) \ , \quad \partial_t w + \boldsymbol{u} \cdot \boldsymbol{\nabla} w = -\partial_z p_{nh} + F_w \tag{2}$$

where $\boldsymbol{u} = (u, v)$ is the horizontal velocity, the dots stand for all other terms of pe_realistic, and F_w is the friction of w.

The equation for w carries neither the buoyancy nor p_h nor p_s. The full equation would be $\partial_t w + \boldsymbol{u} \cdot \boldsymbol{\nabla} w = -\partial_z p - g\rho/\rho_0 + F_w$. p_s does not depend on z. p_h is integrated downward from the surface with

$$p_h(k) = p_h(k+1) + \frac{g}{2\rho_0} \left(\rho(k+1) + \rho(k)\right) \Delta z^w_k \tag{3}$$

which is the integration of pe_realistic. Eq. (3) gives at the W point k the vertical gradient

$$\frac{p_h(k+1) - p_h(k)}{\Delta z^w_k} = -\frac{g}{2\rho_0} \left(\rho(k+1) + \rho(k)\right) \tag{4}$$

The right hand side is the buoyancy force averaged to the W point. The vertical gradient of p_h and the buoyancy force therefore cancel exactly, also on a stretched grid. Only p_{nh} is left in the equation for w.

With $p_{nh} = 0$ and w taken from continuity the model is pe_realistic. The attribute `enable_nonhydrostatic` (default on) selects the non-hydrostatic model. With it off the time step is that of pe_realistic to the last bit. This is tested.

Advection and friction of w

The advection of w is the energy-conserving second-order form of the momentum advection of pe_realistic. The transports through the faces of a T cell are

$$U_i = \Delta y^t \Delta z^t u_i \ , \quad V_j = \Delta x^t \Delta z^t v_j \ , \quad W_k = A^t w_k \tag{5}$$

with the masks of the faces. The transport through a face of the W cell k is the mean of the transports through the same face of the T cells k and $k+1$. The fluxes of w are

$$f^x_i = -\frac{1}{4} \left(U_{i,k} + U_{i,k+1}\right) \left(w_i + w_{i+1}\right) \ , \quad f^z_k = -\frac{1}{4} \left(W_k + W_{k+1}\right) \left(w_k + w_{k+1}\right) \tag{6}$$

and f^y correspondingly with V. f^z_k sits at the T point $k+1$. The advective tendency of w_k is the divergence of these fluxes divided by the volume $A^t \Delta z^w_k$ of the W cell.

The divergence of the transports through the W cell k is the mean of the divergences of the T cells k and $k+1$. It vanishes when the velocity is free of divergence. The advection then conserves w^2. On a uniform grid it is the advection of w of nh_simple.

The equation for w has no Coriolis term, as in pyOM2.

The horizontal friction of w uses the coefficients `Ahbi` and `Ah` of the horizontal friction of u and v. It uses the operator at the T points of pe_realistic, since w sits at the T points in the horizontal. For w this operator is exact. A flux exists only between two wet W cells.

The vertical friction of w has the flux at the T point k between the W cells $k-1$ and k,

$$F_k = A_k \frac{w_k - w_{k-1}}{\Delta z_k^t} \tag{7}$$

The tendency of w_k is $(F_{k+1} - F_k)/\Delta z_k^w$. The roles of Δz^t and Δz^w are thus swapped with respect to the vertical friction of u and v. A flux exists only between two wet W cells.

The coefficient A_k follows the vertical friction of u and v. It is the constant `A_v` with `vmix_scheme` explicit or implicit. With the TKE closure it is the viscosity of the closure averaged from the two W points to the T point. The step is a forward step for `vmix_scheme` explicit and an implicit step otherwise. The implicit step is the tridiagonal solve of pe_realistic on the W grid. The energy that the friction of w removes is not given to the TKE closure.

Projection

The velocities are first advanced with all terms except the gradient of p_{nh}. This gives u^*, v^* and w^*. The surface pressure p_s is already part of u^* and v^*. The vertical friction is also part of them.

The vertical velocity at the surface face is taken from continuity as in pe_realistic,

$$w_{surf} = -\frac{1}{A^t} \sum_k \left(U_i - U_{i-1} + V_j - V_{j-1} \right) \tag{8}$$

with u^* and v^* in the transports and the sum over the wet cells of the column. With the free surface w_{surf} is the change of the surface elevation in time. With the rigid lid it is zero to the tolerance of the surface pressure solver.

The divergence of the T cell is

$$D = \frac{1}{V^t} \left(U_i - U_{i-1} + V_j - V_{j-1} + W_k - W_{k-1} \right) \tag{9}$$

The velocity is corrected by

$$u_i = u_i^* - \Delta t \frac{p_{i+1} - p_i}{\Delta x_i^u} \ , \quad v_j = v_j^* - \Delta t \frac{p_{j+1} - p_j}{\Delta y_j^u} \ , \quad w_k = w_k^* - \Delta t \frac{p_{k+1} - p_k}{\Delta z_k^w} \tag{10}$$

with $p = p_{nh}$, on the wet faces and the interior faces in z. The surface face is closed in Eq. (10). w_{surf} therefore stays as it is.

The pressure p_{nh} makes D of the corrected velocity vanish in every wet cell. Inserting Eq. (10) into $D = 0$ gives

$$\begin{aligned} a_i^x \left(p_{i+1} - p_i \right) - a_{i-1}^x \left(p_i - p_{i-1} \right) + a_j^y \left(p_{j+1} - p_j \right) - a_{j-1}^y \left(p_j - p_{j-1} \right) & \\ + a_k^z \left(p_{k+1} - p_k \right) - a_{k-1}^z \left(p_k - p_{k-1} \right) &= \frac{V^t D^*}{\Delta t} \end{aligned} \tag{11}$$

with D^* the divergence of u^*, v^*, w^* and w_{surf}. The coefficients are

$$a_i^x = \frac{\Delta y^t \Delta z^t}{\Delta x_i^u} \ , \quad a_j^y = \frac{\Delta x^t \Delta z^t}{\Delta y_j^u} \ , \quad a_k^z = \frac{A^t}{\Delta z_k^w} \tag{12}$$

on the wet faces and zero on the closed faces. Eq. (11) is the volume times the divergence of the gradient of p_{nh}.

The coefficient of a face is the same in the equations of the two cells next to it. The matrix of Eq. (11) is therefore symmetric, also on a stretched grid with topography. This is tested. Without the volume factor the matrix would not be symmetric on a stretched grid.

All faces at the boundary of the wet region are closed in Eq. (11). A solution exists only if the right hand side summed over the wet cells vanishes. Summed over a column, the transports through the inner faces cancel. What is left is the sum of the horizontal transports of the column and $A^t w_{surf}$. With Eq. (8) this sum is zero. The right hand side thus sums to zero in every column, and Eq. (11) has a solution.

Eq. (11) fixes p_{nh} only up to a constant. After every iteration the mean of p_{nh} over the wet cells is set to zero, as in Oceananigans. The same is done with the residual.

Solver

Eq. (11) is solved with the method of conjugate gradients with a preconditioner. The solver is written for JAX and MPI in the same way as the solver for p_s of pe_simple. The first guess is p_{nh} of the previous time step.

The preconditioner solves the vertical part of Eq. (11) exactly in every column. It drops the horizontal coefficients a^x and a^y. The vertical part is the tridiagonal system

$$-a_{k-1}^z z_{k-1} + (1+\epsilon)\left(a_k^z + a_{k-1}^z\right) z_k - a_k^z z_{k+1} = r_k \tag{13}$$

for the preconditioned residual z with the residual r. It is solved with the Thomas algorithm. The factor $1+\epsilon$ with $\epsilon = 0.01$ makes the system of each column regular, as in Oceananigans. In a dry cell and in a wet cell without a wet cell above or below, the preconditioner is the identity. The preconditioner is that of Marshall et al. (1997).

The preconditioner suits grids with $\Delta z \ll \Delta x$. There the vertical coefficients a^z are much larger than the horizontal ones. In the test configuration they are larger by a factor of $3 \cdot 10^4$ to $2 \cdot 10^7$. The preconditioner reduces the number of iterations there from 1000 to 3500 to 50 to 150 per time step. For $\Delta z \approx \Delta x$ it helps less.

The residual r is the right hand side minus the left hand side of Eq. (11). The divergence of the velocity corrected with the current p_{nh} is $\Delta t\, r / V^t$. The iteration stops when $\Delta t \max |r| / V^t$ is smaller than `epsilon_nh` (default $10^{-12}\,\mathrm{s}^{-1}$). `epsilon_nh` is thus the largest divergence that the velocity keeps. The iteration also stops after `max_iterations_nh` iterations (default `max_iterations`).

The stopping condition is met by the vertical structure of p_{nh}. The horizontal structure of p_{nh} is less well determined when $\Delta z \ll \Delta x$. It still changes with the round-off of the

sums in the solver. Runs with different domain decompositions therefore differ by about 10^{-8} relative with the default `epsilon_nh`. With `epsilon_nh` $= 10^{-16}$ they differ by about 10^{-10}.

Tracers

The tracers are advected with u, v and w of the end of the previous time step, as in pe_realistic. In pe_realistic w follows from continuity and the velocity is free of divergence to round-off. In nh_realistic the velocity is free of divergence to `epsilon_nh`. A uniform tracer therefore stays uniform only to `epsilon_nh` Δt per time step. The content of a tracer is still conserved to round-off, since the advection is in flux form. The WENO scheme of pe_realistic works in nh_realistic unchanged. For a uniform tracer it gives the same tendency as the mean of the two cells, the tracer times the divergence that the solve leaves.

Time step

The order of the calls in a time step of the split-explicit scheme is

1. diffusivities of the TKE closure (`set_tke_diffusivities`)
2. momentum tendencies, Coriolis force and hydrostatic pressure (`momentum_tendency_expl`)
3. advective tendency of w (`w_tendency`)
4. forcing of the configuration (`set_forcing`)
5. horizontal friction of w and Adams–Bashforth update of w
6. bottom friction
7. external mode with sub cycling (`solve_expl_surface`)
8. vertical friction of u, v and of w
9. surface vertical velocity and the solve for p_{nh} (`nh_projection`)
10. tracer advection and biharmonic tracer mixing, or the advection outside the Adams–Bashforth step
11. Adams–Bashforth update of the tracers
12. isoneutral mixing
13. vertical mixing of the tracers
14. density

15. TKE step (`integrate_tke`)

The order of the calls in a time step of the rigid lid and of the implicit free surface is

1. diffusivities of the TKE closure
2. tracer advection and biharmonic tracer mixing
3. momentum tendencies and hydrostatic pressure (`momentum_tendency_impl`)
4. advective tendency of w (`w_tendency`)
5. forcing of the configuration (`set_forcing`)
6. biharmonic and harmonic friction of u, v and of w
7. Adams–Bashforth update of u, v and of w
8. direct forcing of the configuration (`set_forcing_direct`)
9. surface pressure and its gradient (`solve_for_pressure`)
10. vertical friction of u, v and of w
11. bottom friction
12. surface vertical velocity and the solve for p_{nh} (`nh_projection`)
13. Adams–Bashforth update of the tracers, or the advection outside the Adams–Bashforth step
14. isoneutral mixing
15. vertical mixing of the tracers
16. density
17. TKE step

In both schemes the steps are those of pe_realistic. The steps for w are added. The solve for p_{nh} replaces the vertical velocity from continuity.

The forcing of the configuration may add to the tendency `dw` at the level `tau`, as it does for `du` and `dv`. The tendency then enters the Adams–Bashforth step.

The solve for p_{nh} comes after the vertical friction and the bottom friction. The velocity at the end of the time step is therefore free of divergence. The surface pressure is solved before. It removes the depth-averaged divergence, which is the mode with the slowest convergence in Eq. (11).

The restart needs the fields of pe_realistic and in addition `dw` and `p_nh`. Without `p_nh` the solver starts from another first guess. The run is then no longer identical to the last bit.

Model parameters and options

The parameters and options of pe_realistic are also those of nh_realistic. The tables list only the differences.

Required parameters

Attribute	Unit	Meaning
`dxt`, `dyt`	m	widths of the cells in x and y, as in pe_realistic but in metres only

`coord_degree` must be off.

Options with defaults

Attribute	Default	Meaning
`enable_nonhydrostatic`	on	prognostic w and the solve for p_{nh}. Off gives pe_realistic
`epsilon_nh`	10^{-12}	largest divergence (s^{-1}) that the solve for p_{nh} leaves
`max_iterations_nh`	`max_iterations`	largest number of iterations of the solve for p_{nh}
`Ah`, `Ahbi`	unset	as in pe_realistic, also acting on w
`A_v`	0	as in pe_realistic, also acting on w
`weno_advection`, `weno_ref`	off, 1	WENO tracer advection as in pe_realistic. It needs `halo_points` = 3

The number of iterations of the last solve for p_{nh} is `congr_itts_nh`. The divergence it leaves is `estimated_error_nh`.

Snapshot diagnostics

The snapshot classes `snap_nc4` and `snap_raw` are those of pe_realistic. `snap_nc4` also writes p_{nh} under the name `p_nh`. w is written on the W points as in pe_realistic. It is now the prognostic w.

Limits

- The horizontal coordinates must be Cartesian, see the grid above.
- The equation for w has no Coriolis term.

- The energy that the friction of w removes is not given to the TKE closure.
- With the free surface w_{surf} does not enter the advection of w, since the surface face is closed in the equation for w. The advection of w then does not conserve w^2 exactly in the uppermost W cell.
- The preconditioner suits $\Delta z \ll \Delta x$. For $\Delta z \approx \Delta x$ the number of iterations grows. A preconditioner with the FFT of a uniform grid, as in Oceananigans, is not available.
- Runs with different domain decompositions are not identical to the last bit, see the solver above.
- The LES closures of nh_simple are not part of nh_realistic.